

Product Reference Manual (V1.0)

Description

The DevLab I²C Level Shifter Module is a compact and efficient solution designed to enable reliable bidirectional communication between devices operating at different voltage levels, specifically 3.3V and 5V systems. This module simplifies hardware integration by eliminating the need for external level-shifting circuits, making it ideal for rapid prototyping and embedded development.

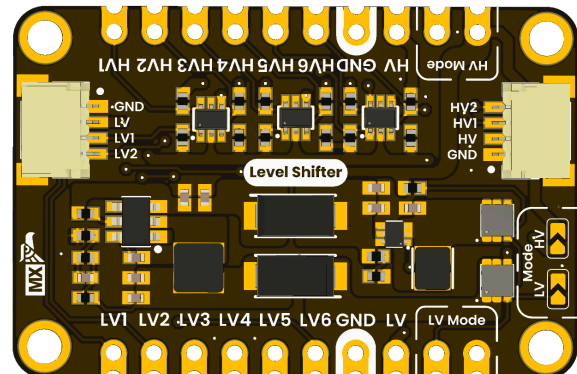
It incorporates integrated voltage regulation and standardized connectors compatible with Qwiic and STEMMA QT, allowing seamless connection and daisy-chaining of sensors and peripherals. The module is optimized for I²C communication, ensuring stable signal translation while maintaining signal integrity across mixed-voltage environments.

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards.

This design guarantees efficient prototyping, accessibility, and simplicity of integration across a variety of devices and modules.



Key Features

- Bidirectional Level Shifting Supports automatic voltage translation between 3.3V and 5V I²C devices.
- The voltage mode is configurable via external castellated headers and solder pads, allowing operation in
 - Buck mode (5V → 3.3V)
 - Boost mode (3.3V → 5V)
- Qwiic / STEMMA QT Compatibility Standard 4-pin JST SH connectors enable quick and solderless integration.
- Integrated Power Regulation Built-in voltage conversion removes the need for external regulators.
- Compact and Breadboard-Friendly Design Optimized footprint for prototyping and embedded applications.

Hardware Features

- Operating Voltage Range: 3.3V / 5V selectable
- Communication Interface: I²C (bidirectional)
- Connectors:
 - 2 × JST SH 4-pin (Qwiic/STEMMA QT compatible)
 - Header pins for direct connection
- Level Shifting Method: MOSFET-based bidirectional translation
- Configuration: External castellated headers and solder pads for voltage selection
- Integrated Voltage Regulator (buck/boost support)
- Pull-up resistors included for I²C lines
- Compact PCB design for DevLab ecosystem

Software Support

- No dedicated driver required (transparent hardware-level operation)
- Fully compatible with common sensors and peripherals using I²C protocol

Applications

- Mixed-voltage I²C communication (3.3V ↔ 5V systems)
- Rapid prototyping with modular sensor ecosystems
- IoT development platforms (ESP32, RP2040, Arduino, etc.)
- Sensor interfacing and signal conditioning
- Educational and laboratory environments
- Expansion of embedded systems with heterogeneous voltage domains

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1. The Board

1.1 Accessories

2 Ratings

2.1 Recommended Operating Conditions

Parameter	Description	Min	Typ	Max	Unit
VIN	Input supply voltage for module operation	2.5	3.3 / 5.0	5.5	V
VLV	Low-voltage rail operating range	1.8	3.3	5.0	V
VHV	High-voltage rail operating range	3.3	5.0	V	
IOUT-LV	Maximum output current for LV rail	-	-	1	A
IOUT-HV	Maximum output current for HV rail	-	-	1	A
VIH	Input high-level voltage for I ² C logic	0.7 × VCC	-	-	V
VIL	Input low-level voltage for I ² C logic	-	-	0.3 × VCC	V
VOH	Output high-level voltage for I ² C logic	0.8 × VCC	-	-	V
VOL	Output low-level voltage for I ² C logic	-	-	0.2 × VCC	V
fI2C	Supported I ² C bus frequency	0	100	400	kHz
TAMB	Recommended operating ambient temperature	-20	25	85	°C
RPULLUP	Integrated pull-up resistor value	-	10	-	kΩ
ESD	ESD protection level (HBM typical)	-	±2	-	kV

Note 1: Maximum output current depends on thermal dissipation, PCB layout, and input voltage conditions.

Note 2: The module integrates bidirectional MOSFET-based level shifters optimized for open-drain I²C communication.

Note 3: Qwiic/STEMMA QT connectors share the same I²C bus signals (VCC, GND, SDA, SCL) for daisy-chain operation.

Note 4: Operating voltage ranges may vary depending on external power configuration and selected LV/HV mode.

3. Functional Overview

The DevLab I²C Level Shifter Module operates as a transparent interface between devices that use different logic voltage levels, specifically **3.3V and 5V systems**, enabling safe and reliable bidirectional communication over the I²C bus.

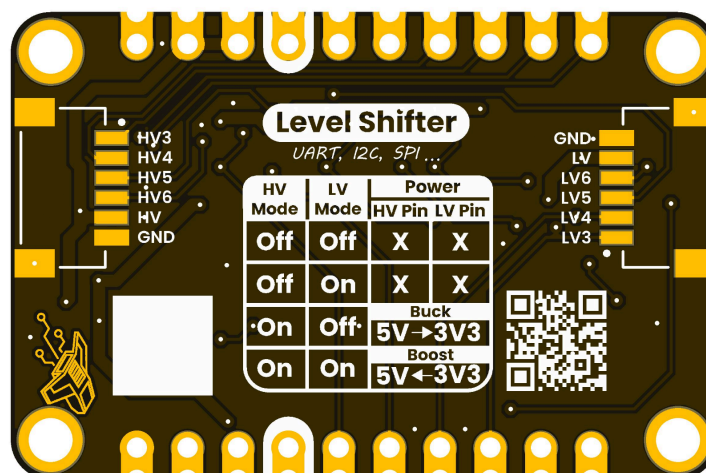
Level Shifter

A level shifter is an electronic circuit that translates signal voltages between different logic levels, enabling communication between devices that operate at incompatible voltage domains (e.g., 3.3V and 5V systems).

At its core, the module implements a **MOSFET-based bidirectional level-shifting architecture**, which allows both the **SDA (data)** and **SCL (clock)** lines to dynamically adapt their voltage levels depending on the direction of communication. This ensures that signals originating from a lower-voltage device (e.g., 3.3V microcontroller) can be correctly interpreted by a higher-voltage device (5V sensor) and vice versa without requiring software intervention.

The module integrates an onboard **voltage regulator**, configurable through a **External Castellated Headers and Solder Pads**, allowing the user to define the operating mode:

- **Buck mode (5V → 3.3V):** Used when the primary system operates at 5V and peripherals require 3.3V.
- **Boost mode (3.3V → 5V):** Used when the host system is 3.3V and needs to interface with 5V devices.



This flexibility allows the module to adapt to a wide range of embedded platforms without redesigning the power stage.

To simplify integration, the module includes **dual Qwiic® / STEMMA QT connectors**, enabling plug-and-play connectivity and daisy-chaining of multiple I²C devices. The presence

of **integrated pull-up resistors** ensures proper bus operation, reducing the need for external components.

From a system perspective, the module behaves as a **passive communication bridge**, meaning it does not alter data content or protocol timing. Instead, it preserves the electrical integrity of the I²C signals while maintaining compatibility with standard I²C speeds.

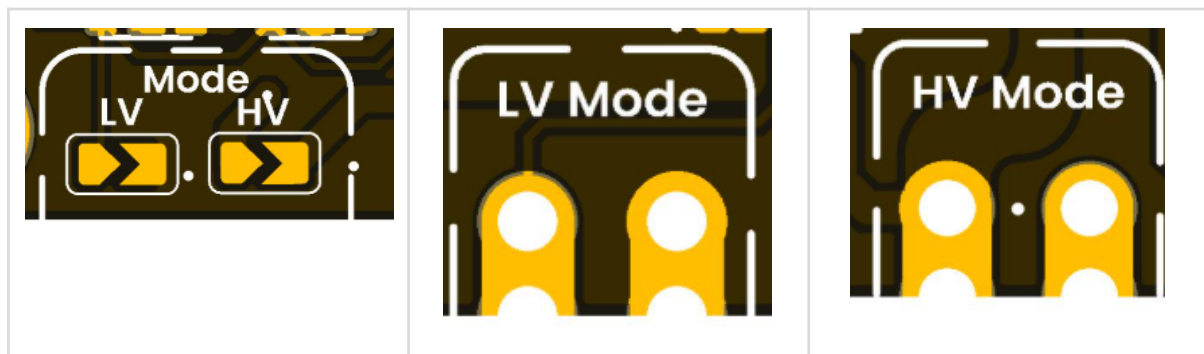
Overall, the module enhances system interoperability by:

- Eliminating voltage mismatch issues
- Reducing external circuitry
- Enabling modular and scalable I²C architectures

This makes it particularly suitable for **rapid prototyping, multi-device sensor networks, and mixed-voltage embedded systems** within the DevLab ecosystem.

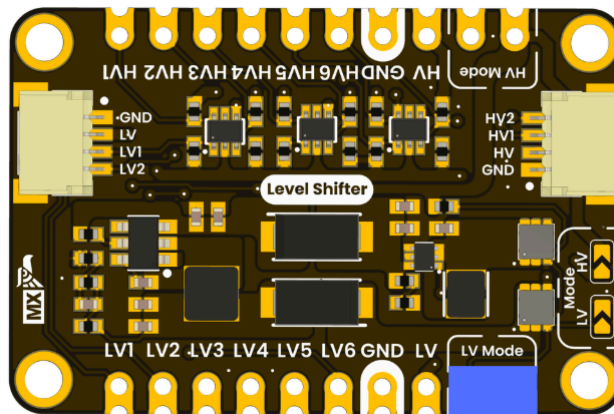
Mode Configuration (Boost / Buck)

Mode configuration is defined through a hardware selection mechanism implemented using header pins or solder pads. The user must establish a physical connection (bridge) to select the desired operating mode, either boost or buck. This configuration determines the direction of voltage conversion and must be completed prior to powering the system. The use of solder pads or jumpers ensures a stable and fixed configuration suitable for both prototyping and production environments.



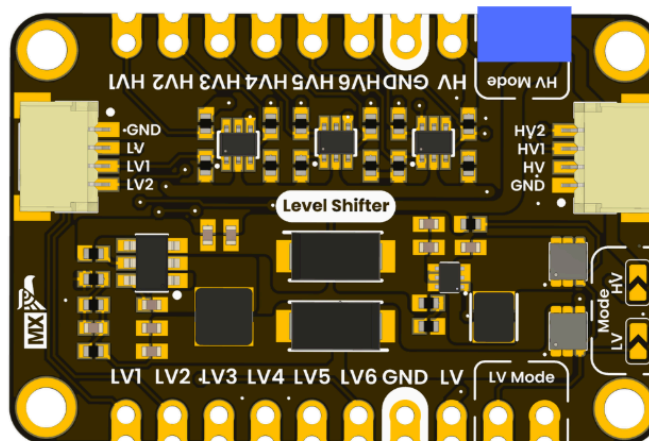
Boost Mode (3.3V → 5V)

Boost mode is defined as the operating condition in which the module receives a 3.3V input and internally elevates this voltage to 5V to supply the high-voltage domain. This configuration enables low-voltage microcontrollers, such as ESP32 or RP2040, to interface with peripherals that require 5V logic levels. Under this condition, the low-voltage side operates at 3.3V while the high-voltage side is generated by the internal boost regulator. The available output current is limited by the characteristics of the boost stage, and overall efficiency depends on load conditions and input stability.



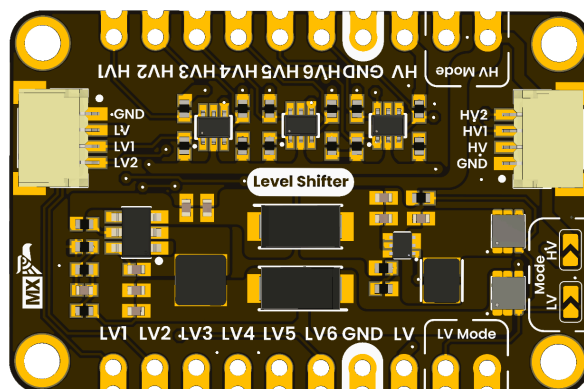
Buck Mode (5V → 3.3V)

Buck mode is defined as the operating condition in which the module accepts a 5V input and regulates it down to 3.3V for use in the low-voltage domain. This configuration is intended for systems where a 5V controller must interface with or supply modern 3.3V devices. In this mode, the high-voltage side remains at 5V while the low-voltage side is regulated to 3.3V. The regulator is optimized for low to moderate current consumption, making it suitable for sensor-based and embedded applications.



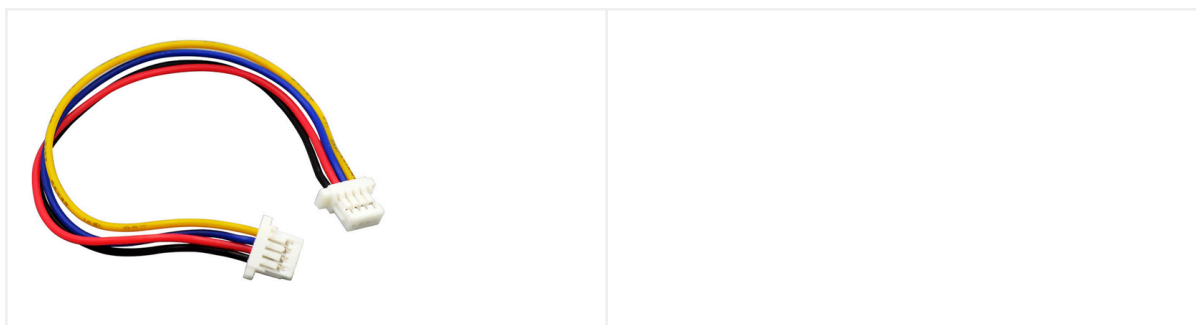
Inline (In-Line) Operation

Inline operation refers to the module's ability to be inserted directly between two I²C domains, functioning as a transparent electrical bridge. In this mode of use, the module does not require firmware configuration and does not alter the structure, timing, or content of transmitted data. It preserves the integrity of the I²C protocol, allowing seamless communication between devices operating at different voltage levels without additional software overhead.



JST Connector (Qwiic / STEMMA QT)

The module incorporates JST SH 4-pin connectors, which define a standardized interface for I²C communication and power distribution. These connectors carry VCC, GND, SDA, and SCL signals, enabling direct plug-and-play integration. Their design supports daisy-chaining of multiple modules, reducing wiring complexity and minimizing connection errors while ensuring compatibility with Qwiic and STEMMA QT ecosystems.

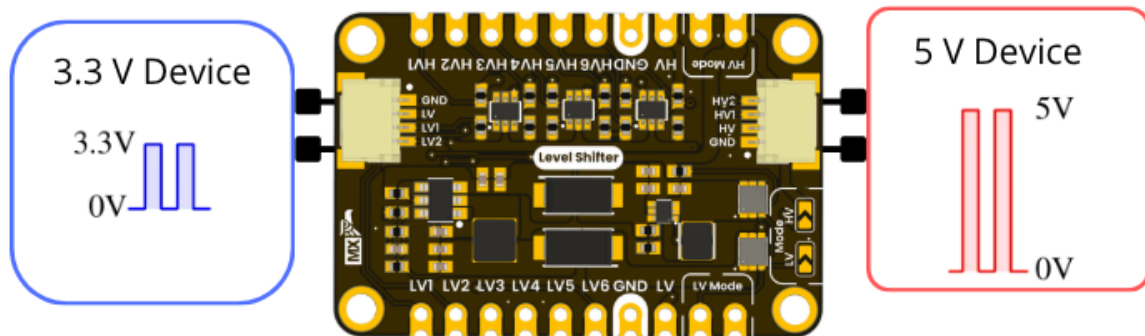


Operational Considerations

Operational considerations refer to the conditions required to ensure stable and reliable system performance. Proper selection of the operating mode is essential, as incorrect configuration may lead to incompatible voltage levels. When multiple devices are connected, total bus capacitance must be evaluated, as excessive capacitance can degrade signal integrity. The interaction of pull-up resistors across modules should also be considered to avoid excessive current draw. Additionally, a common ground reference between both voltage domains must be maintained, and standard I²C speeds, typically around 100 kHz in complex systems, are recommended to ensure reliable communication.

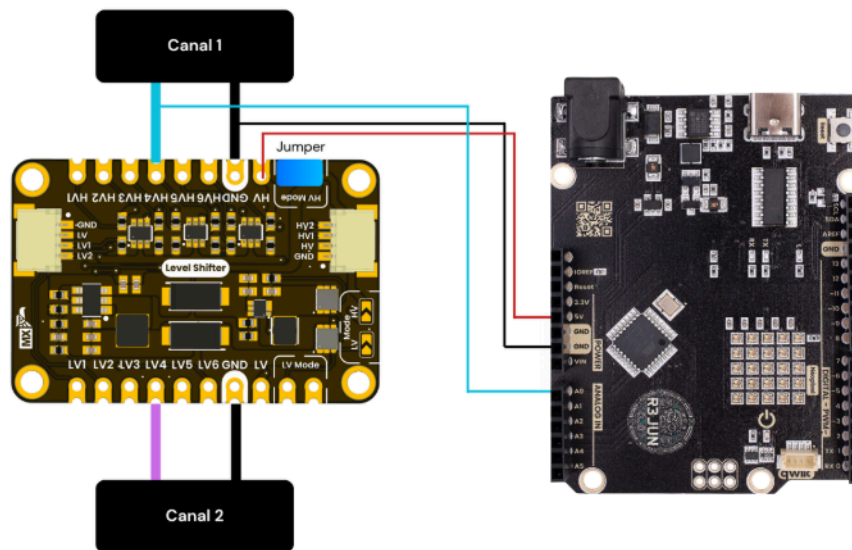
Use Cases

Use cases define the typical application scenarios for the module. These include mixed-voltage sensor integration, where devices with different logic levels must communicate; modular DevLab prototyping systems, where rapid and flexible hardware assembly is required; IoT nodes incorporating heterogeneous peripherals; educational platforms for demonstrating embedded system concepts; and validation environments where quick hardware interfacing is necessary.



Signal and Multi-Device Operation

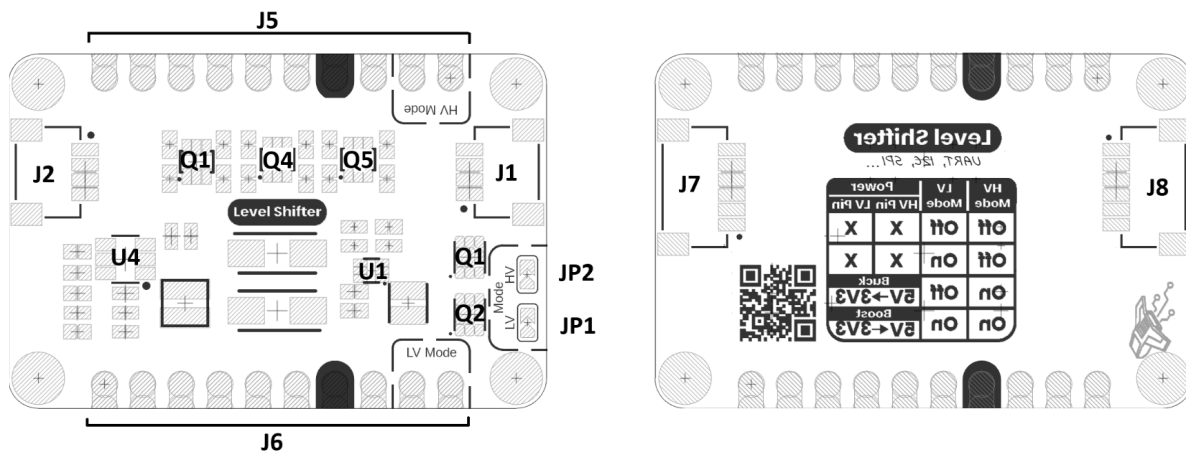
Signal and multi-device operation describe the module's ability to support multiple I²C devices across both voltage domains. Each side of the module behaves as an independent bus segment while maintaining communication through the level shifting stage. Devices connected within the same segment must have unique I²C addresses to avoid conflicts. The module maintains compatibility with standard I²C features, including clock stretching, ensuring reliable operation in complex multi-device configurations.



Power Characteristics (Buck Mode Reference)

The power characteristics in buck mode are defined by the behavior of the internal regulator under load conditions. The minimum input current is determined by the connected devices and is typically low in sensor-based applications. The regulator design limits the maximum output current, making the module suitable for low- to medium-power loads. It is not intended for high-power applications such as motors or large displays, but rather for efficient operation within embedded systems and distributed sensor networks.

3.2 Board Topology



Views of Topology

Table 3.2.1 - Components Overview

Ref.	Description
U1	Boost converter circuit based on TPS61023 for LV power rail generation
U4	Buck converter circuit based on TPS54302DDCT for HV power rail generation
Q1, Q4, Q5	Bidirectional I ² C level shifter circuits based on 2N7002KDW MOSFETs
Q2, Q3	Power switching MOSFET stage for voltage rail selection/control
J1, J2	Qwiic/STEMMA QT connectors (JST 1.0 mm pitch) for I ² C interface
J5, J6	Castellated 1x10 pin headers (2.54 mm pitch) for external module integration
J7, J8	1x6 pin headers/connectors for HV/LV mode configuration and external interfacing
JP1	LV Mode solder jumper selector
JP2	HV Mode solder jumper selector

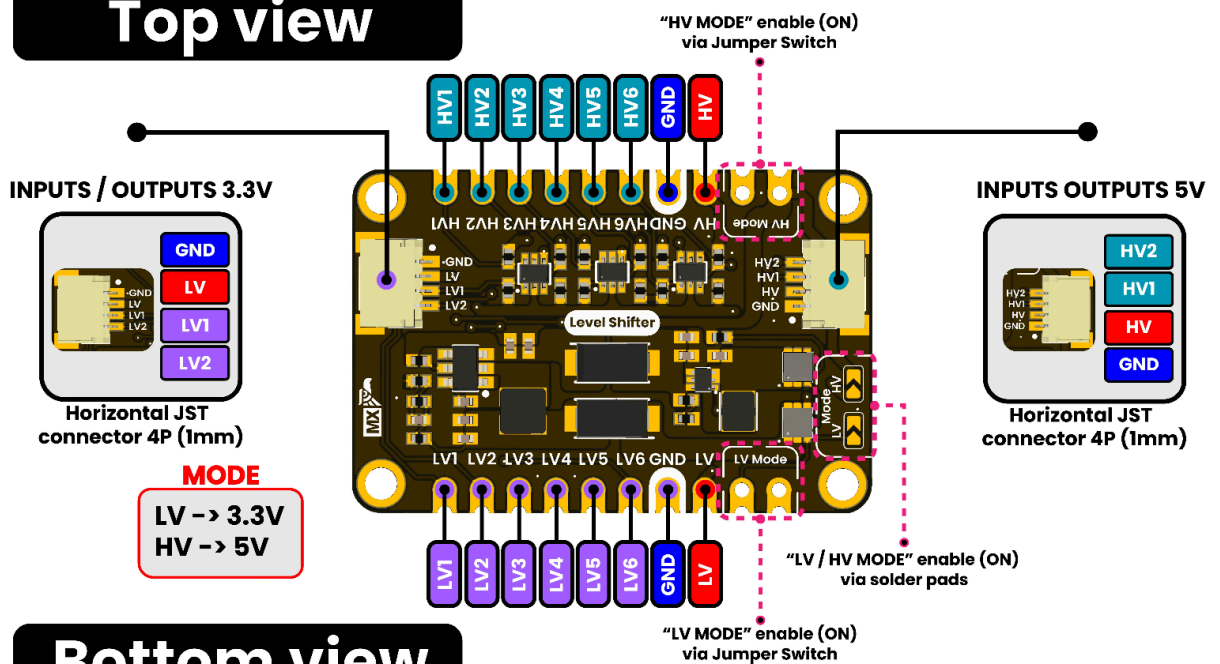
4 Connectors & Pinouts

4.1 General Pinout

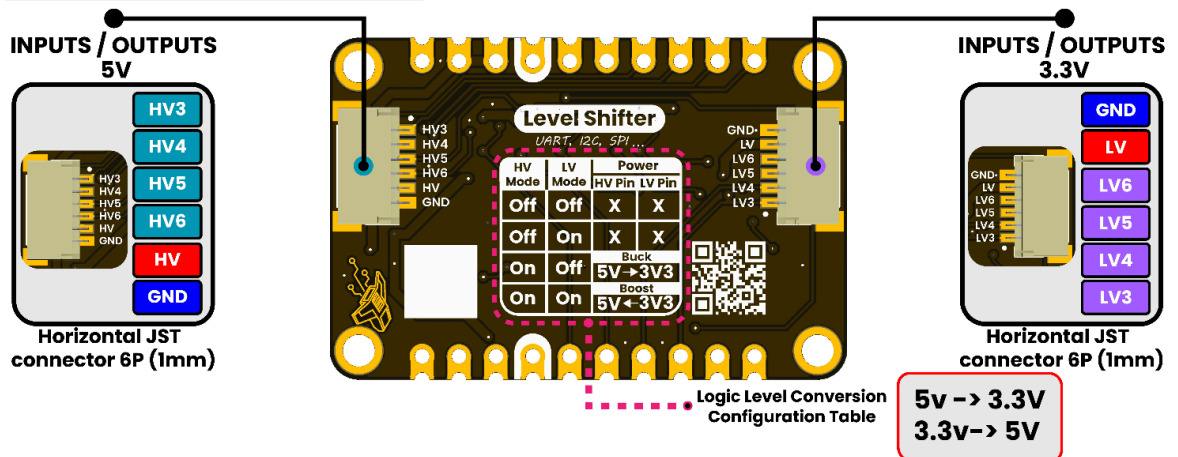
PINOUT

DevLab: I2C Qwiic Converter Module

Top view



Bottom view



DevLab I2C Converter Module General Pinout

4.2 Pinout General Description

Pin	Label	Description	Typical Use
1	LV1	Low-voltage logic channel 1 (3.3 V side)	I ² C SDA, GPIO signals
2	LV2	Low-voltage logic channel 2 (3.3 V side)	I ² C SCL, GPIO signals
3	LV3	Low-voltage logic channel 3 (3.3 V side)	UART, SPI, GPIO
4	LV4	Low-voltage logic channel 4 (3.3 V side)	UART, SPI, GPIO
5	LV5	Low-voltage logic channel 5 (3.3 V side)	Digital I/O expansion
6	LV6	Low-voltage logic channel 6 (3.3 V side)	Digital I/O expansion
7	GND	Common ground reference	System ground connection
8	LV	Low-voltage power rail	3.3 V supply input/output
9	HV	High-voltage power rail	5 V supply input/output
10	HV1	High-voltage logic channel 1 (5 V side)	I ² C SDA, GPIO signals
11	HV2	High-voltage logic channel 2 (5 V side)	I ² C SCL, GPIO signals
12	HV3	High-voltage logic channel 3 (5 V side)	UART, SPI, GPIO
13	HV4	High-voltage logic channel 4 (5 V side)	UART, SPI, GPIO
14	HV5	High-voltage logic channel 5 (5 V side)	Digital I/O expansion
15	HV6	High-voltage logic channel 6 (5 V side)	Digital I/O expansion

16	Qwiic / STEMMA QT Connector	4-pin JST-SH I ² C connector (3.3 V side)	Sensor and peripheral connection
17	HV JST Connector	4-pin JST connector (5 V side)	5 V I ² C device connection
18	LV Mode Jumper	Enables low-voltage operating mode	3.3 V level-shifting configuration
19	HV Mode Jumper	Enables high-voltage operating mode	5 V level-shifting configuration
20	LV/HV Mode Pads	Solder-selectable configuration pads	Buck/Boost and voltage mode selection

Signal Groups

Group	Description
LV1–LV6	Low-voltage signal channels referenced to the LV supply rail (typically 3.3 V).
HV1–HV6	High-voltage signal channels referenced to the HV supply rail (typically 5 V).
LV	Low-voltage power rail used by the Qwiic/STEMMA QT interface and low-voltage side circuitry.
HV	High-voltage power rail used by external 5 V devices and high-voltage side circuitry.
GND	Common ground shared by both voltage domains.

Connector Pinout

Qwiic / STEMMA QT Connector (3.3 V Side)

Pin	Signal
1	GND
2	LV
3	LV1
4	LV2

HV JST Connector (5 V Side)

Pin	Signal
1	HV2
2	HV1
3	HV
4	GND

HV JST Connector (6-Pin Version)

Pin	Signal
1	HV3
2	HV4
3	HV5
4	HV6
5	HV
6	GND

LV JST Connector (6-Pin Version)

Pin	Signal
1	GND
2	LV
3	LV6
4	LV5
5	LV4
6	LV3

5 Board Operations

The DevLab I²C Level Shifter Module is designed to operate as a transparent bidirectional voltage translation interface between two independent I²C voltage domains. Proper board operation requires correct voltage mode selection, appropriate power connections, and valid I²C bus topology.

5.1 Getting Started

To begin operation, connect the module between the host controller and the target I²C peripheral devices. The module automatically translates SDA and SCL voltage levels between the low-voltage (LV) and high-voltage (HV) domains.

Typical supported platforms include:

- ESP32 series
- RP2040 / RP2350
- Arduino UNO / Mega
- STM32 platforms
- Sensor modules using Qwiic® or STEMMA QT interfaces

The board supports plug-and-play operation and does not require firmware configuration or dedicated drivers.

5.2 Mode Selection

The module supports two operating configurations:

Boost Mode (3.3V → 5V)	Buck Mode (5V → 3.3V)
This mode is used when the main controller operates at 3.3V and communication with 5V peripherals is required. Operation: • LV side operates at 3.3V • Internal boost regulator generates the 5V HV rail • Recommended for ESP32, RP2040, and other 3.3V systems	This mode is used when the host system operates at 5V and peripherals require 3.3V logic levels. Operation: • HV side remains at 5V • Internal regulator generates the 3.3V LV rail • Suitable for Arduino UNO and 5V embedded platforms Mode selection is configured using the dedicated solder jumpers or hardware selection pads available on the module. The

	configuration must be completed before powering the board.
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5.3 Power Connections

The board distributes power through both header pins and JST SH connectors.

Typical Power Signals

Signal	Description
HV	High-voltage rail
LV	Low-voltage rail
GND	Common ground reference
SDA	I ² C data line
SCL	I ² C clock line

A common ground connection between all devices is mandatory for stable communication.

5.4 I²C Communication

The module functions as a passive inline I²C bridge. It preserves:

- I²C timing
- Clock stretching
- Bidirectional communication
- Standard protocol behavior

No protocol conversion or buffering is performed.

The integrated MOSFET-based level-shifting architecture dynamically adapts signal levels according to communication direction.

5.5 Multi-Device Operation

Multiple I²C devices may be connected simultaneously using the Qwiic® / STEMMA QT daisy-chain topology.

Recommendations:

- Ensure all devices use unique I²C addresses
- Minimize excessive cable length

- Avoid excessive bus capacitance
- Maintain stable supply voltage

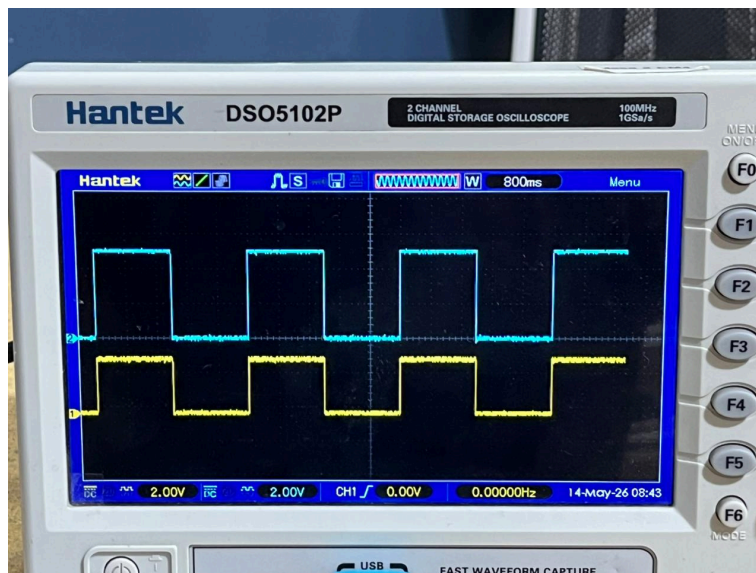
For complex systems, standard-mode I²C operation (100 kHz) is recommended to maximize reliability.

5.6 Typical Connection Example

3.3V Controller to 5V Peripheral

Example:

- ESP32 → DevLab Level Shifter → 5V sensor



Configuration:

- Select Boost Mode
- Connect ESP32 to LV side
- Connect 5V device to HV side

5V Controller to 3.3V Sensor

Example:

- Arduino UNO → DevLab Level Shifter → 3.3V sensor

Configuration:

- Select Buck Mode
- Connect Arduino to HV side
- Connect sensor to LV side

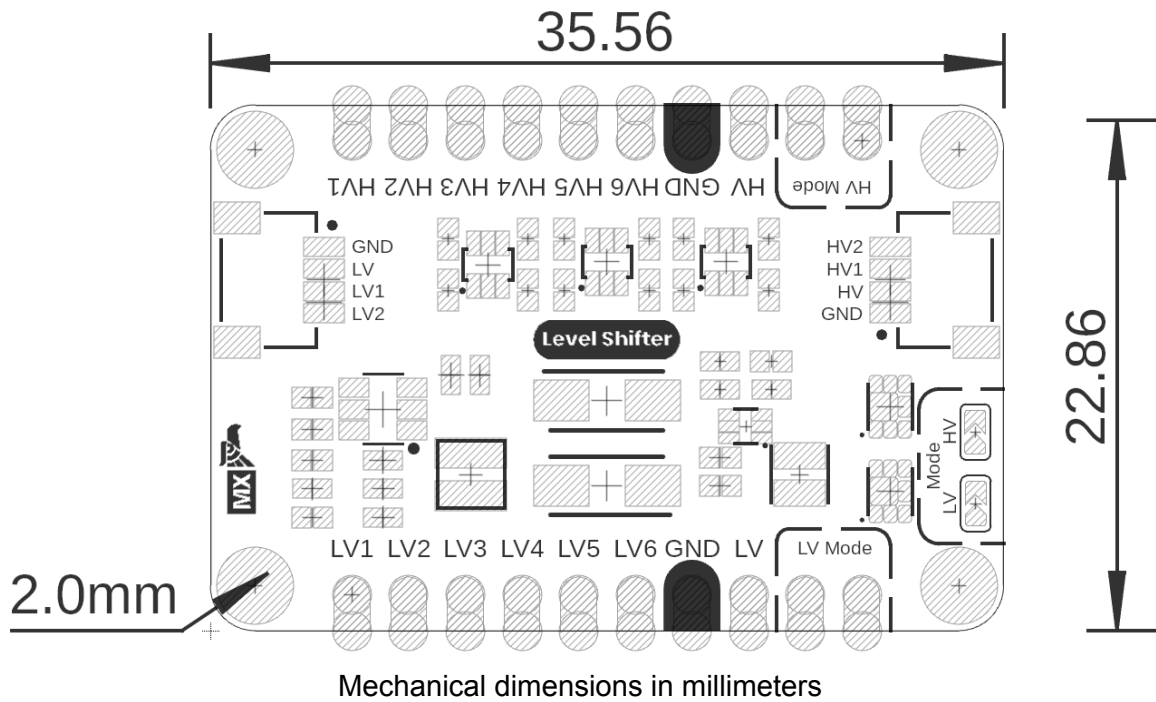
5.7 Operational Notes

For reliable operation:

- Configure voltage mode before power-up
- Do not exceed the regulator's current capabilities
- Avoid connecting high-power loads
- Verify pull-up resistor interactions in large I²C networks
- Ensure correct polarity on all connectors

The module is intended for embedded communication systems, sensor integration, educational platforms, and rapid prototyping applications

6. Mechanical Information



7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://wiki.uelectronics.com/wiki/unit_devlab_i2c_qwiic_converter_module
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_qwiic_converter_module
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_qwiic_converter_module/blob/main/hardware/unit_sch_v_1_0_0_ue0106_i2c_qwiic_converter_module.pdf)

